

Value Change Causation Link Module (VCCM)

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Canonical Definition

Value Change Causation Link Module (VCCM) is the fourth Core Module of the Value Change Standard (VCS) within VFM® — Value Flow Mechanisms Architecture.

VCCM governs the establishment, qualification, and documentation of the relationship between an identified Value Change and the event, condition, activity, mechanism, decision, interaction, transformation, or other factor associated with that change.

The Module does not attempt to become a universal causality framework and does not independently determine scientific, legal, financial, technical, regulatory, or operational causation where those determinations belong to another methodology or authority.

Its function is narrower:

to establish how a governed Value Change is linked to its identified causal basis without treating correlation, sequence, attribution, or assumption as proven causation.

VCCM therefore enables VFM® to answer not only:

What changed?

but also:

To what causal basis is that Value Change legitimately linked, and how strong is that link?

Architectural Position

VCCM is the fourth module within VCS.

It follows:

VCMM — Value Change Magnitude Module

and precedes:

VCRM — Value Change Record Module.

The internal sequence is:

Governed Value State → VCIM — Value Change Identification → VCTM — Value Change Type → VCMM — Value Change Magnitude → VCCM — Value Change Causation Link → VCRM — Value Change Record

VCIM establishes:

that Value Change occurred.

VCTM establishes:

what kind of Value Change occurred.

VCMM establishes:

how large or significant the Value Change was.

VCCM establishes:

what causal basis may legitimately be linked to that Value Change.

Internal Module Flow

Identified Value Change → Classified Value Change → Governed Change Magnitude → Potential Causal Factor Identification → Causal Basis Identification → Temporal Relationship Assessment → Mechanism Relationship Assessment → Alternative Cause Assessment → Multi-Cause Assessment → Evidence & Uncertainty Assessment → Causation Link Determination → Causation Link State → VCRM — Value Change Record Module

Operational Space

VCCM governs the space between:

measured or characterized Value Change

and

the governed recording of that Value Change with its causal relationship.

Its operational space includes:

causal-factor identification; causal-basis identification; causal-event linkage; causal-condition linkage; causal-mechanism linkage; causal-activity linkage; causal-decision linkage; causal-interaction linkage; direct and indirect causal relationships; contributing factors; multi-causal relationships; causal chains; causal dependency; causal evidence; temporal sequence; alternative explanations; confounding factors; causal uncertainty; causal-link strength; causal-link limitation; causation-link state; causation reassessment.

Value Change ≠ Cause

The existence of Value Change does not establish its cause.

For example:

Asset Value: -20%

does not establish whether the change resulted from:

market conditions; physical deterioration; operational failure; regulatory intervention; technological obsolescence; financing conditions; reputational damage; reduced demand; supply disruption; another factor; or several factors acting simultaneously.

VCCM therefore separates:

Observed Value Change

from

Causal Explanation of Value Change.

Causal Factor Identification

The first causation task is to identify candidate factors that may be materially related to the Value Change.

A causal factor may include:

event; action; decision; omission; condition; process; transformation; technological development; physical deterioration; market movement; environmental event; regulatory event; payment; investment; innovation; operational improvement; operational failure; infrastructure change; relationship change; resource change; capability change; information event; external shock; another materially relevant factor.

Identification does not establish causation.

It creates a candidate causal basis for assessment.

Causal Basis

A governed causation link requires a basis explaining why the candidate factor is connected to the Value Change.

The basis may include:

direct observation; controlled comparison; operational evidence; historical evidence; technical analysis; statistical evidence; experimental evidence; simulation evidence; contractual evidence; financial evidence; scientific evidence; expert determination; validated external determination; documented mechanism; another governed evidential basis.

VCCM may consume causal determinations from external systems where appropriate.

It does not need to reproduce them.

Temporal Sequence ≠ Causation

An event occurring before Value Change does not establish that the event caused the change.

Therefore:

Event A → Value Change B

does not automatically mean:

Event A caused Value Change B.

VCCM requires temporal sequence to be treated as one relevant relationship, not as sufficient proof of causation.

Correlation ≠ Causation

Two variables may move together without one causing the other.

VCCM therefore preserves the distinction:

Correlation ≠ Causation

A strong correlation may contribute to a causal assessment.

It cannot automatically substitute for one.

Attribution ≠ Causation

An organization may attribute a Value Change to a business unit, project, technology, person, investment, event, or intervention.

That attribution may be useful for reporting.

It is not automatically a governed causal determination.

VCCM therefore distinguishes:

Assigned Attribution

from

Governed Causation Link.

Direct Causation Link

A Direct Causation Link may be established where a factor has a sufficiently supported direct relationship to Value Change.

Example:

Industrial machine destroyed

→ productive capability eliminated

→ operational value materially reduced.

The causal chain is short and may be strongly observable.

Even here, VCCM requires the causal basis to remain documented.

Indirect Causation Link

Value Change may arise through an indirect causal chain.

Example:

New technology introduced

→ production efficiency increases

→ unit production cost decreases

→ economic operating value increases.

The original technological change may be causally linked to the resulting Value Change through intermediate mechanisms.

VCCM preserves those intermediate relationships where materially relevant.

Causal Chain

Complex Value Change frequently results from a chain rather than one isolated event.

VCCM may represent:

Cause A → Intermediate Effect B → Operational Change C → Value Change D.

Longer causal chains may introduce additional uncertainty.

The existence of an upstream event should therefore not automatically establish every downstream Value Change.

Contributing Cause

A factor may contribute materially to Value Change without independently producing the entire change.

VCCM therefore supports:

Contributing Causal Factor.

Example:

A company's operating value falls because of:

Demand Reduction

Energy Cost Increase

Production Disruption

Financing Cost Increase.

No single factor necessarily explains the complete Value Change.

Multi-Causal Value Change

VCCM explicitly supports multi-causal Value Change.

A governed Value Change may have:

one dominant cause; several contributing causes; interacting causes; sequential causes; reinforcing causes; offsetting causes; unresolved causes.

This is particularly important for:

banking; insurance; investment; industrial systems; infrastructure; complex organizations; digital assets; AI systems; supply chains; mission systems.

Causal Contribution

Where evidence permits, VCCM may determine the contribution of different causal factors.

For example:

Total Operating Value Change: –CHF 10M

Potential governed contribution:

Demand Reduction: –CHF 4M

Production Interruption: –CHF 3M

Energy Cost Increase: –CHF 2M

Financing Change: –CHF 1M.

Such decomposition is legitimate only where the methodology and evidence support it.

VCCM must not manufacture false precision merely to force total Value Change into causal percentages.

Interacting Causes

Causes may interact.

For example:

Supply shortage

alone may create limited Value Change.

Supply shortage + production bottleneck

may create disproportionately larger Value Change.

Therefore:

Combined Causal Effect $\neq$ Sum of Independent Effects

where interaction materially changes the outcome.

VCCM must permit such relationships to remain visible.

Offsetting Causes

Different causal factors may move value in opposite directions.

Example:

Automation: +CHF 5M value effect

Energy Price Increase: –CHF 2M value effect

Net Value Change: +CHF 3M.

The observed net change should not erase the underlying positive and negative causal contributions.

No Single-Cause Requirement

VCCM does not require every Value Change to be reduced to one cause.

Where evidence supports only:

multiple plausible causes

the governed result should remain multi-causal or uncertain.

Methodological integrity is stronger than artificial certainty.

Alternative Cause Assessment

Before a strong causal link is established, materially plausible alternative explanations should be considered.

Alternative causes may:

explain the entire change; explain part of the change; weaken the primary causal hypothesis; reveal a multi-causal relationship;

remain unresolved.

The depth of assessment should be proportional to the significance of the Value Change and intended reliance.

Confounding Factors

A confounding factor may influence both the apparent cause and the Value Change.

VCCM requires material confounding relationships to remain visible where known.

This prevents organizations from incorrectly assigning value creation or destruction to a factor that merely coincided with the actual causal mechanism.

Causal Evidence Strength

Causal links may have different evidential strength.

VCCM may classify a causation link as:

Indeterminate

Possible

Plausible

Supported

Strongly Supported

Established

or another governed classification.

These states must be based upon defined criteria.

They are not universal scientific or legal causation standards.

Causation Link States

A minimum governance model may include:

1. Unlinked

No material causal relationship has yet been established.

2. Candidate Link

A potentially relevant causal factor has been identified.

3. Conditional Link

A causal relationship is supported only under defined assumptions, conditions, or limitations.

4. Supported Link

Available evidence materially supports the causal relationship.

5. Established Link

The causal relationship has been sufficiently established for the defined governance purpose.

6. Contested Link

Material evidence or alternative explanations challenge the current causal determination.

7. Superseded Link

A previous causal determination has been replaced by a better-supported determination.

Causal Link Strength ≠ Causal Contribution

A factor may be strongly established as a cause while accounting for only a small part of the total Value Change.

Conversely, a factor may appear capable of explaining most of the Value Change while the causal evidence remains uncertain.

Therefore:

Causal Link Strength ≠ Magnitude of Causal Contribution

These dimensions should remain separately governed.

Causation and Uncertainty

VCCM requires material uncertainty to remain attached to causal determinations.

Uncertainty may arise from:

incomplete evidence; complex interactions; multiple causes; unavailable counterfactuals; uncertain timing; uncertain mechanisms; model dependence; incomplete observations; confounding factors; changing conditions; limited historical data.

A causal determination should not appear more certain than its basis permits.

Counterfactual Reasoning

Where appropriate, VCCM may use a counterfactual question:

What would likely have happened to the governed value if the candidate causal factor had not occurred?

Counterfactual analysis may strengthen causal assessment.

It may use:

historical comparison; control groups; simulation; scenario analysis; econometric methods; engineering models; operational comparison;

other legitimate methodologies.

VCCM governs the relationship of such analysis to Value Change.

It does not prescribe one universal counterfactual methodology.

External Causation Determinations

VCCM may consume causal findings produced by other competent systems.

Examples include:

engineering root-cause analysis; scientific investigation; accident investigation; financial analysis; forensic accounting; insurance investigation; regulatory determination; judicial determination; validated simulation; operational incident

analysis.

VFM® does not need to replace these systems.

It needs to preserve how their determinations connect to Value Change.

Causal Determination Provenance

Where an external determination is used, VCCM should preserve:

source; determination type; date; scope; applicable object; relevant evidence; limitations; authority or methodology; version where applicable.

This allows the Value Change causation link to remain reviewable.

Causation Reassessment

A causation link may require reassessment when:

new evidence appears; alternative causes emerge; a previous assumption fails; causal contribution changes; external determination changes; new operational data becomes available; a causal mechanism is disproven; Value Change is remeasured; a previous link becomes contested.

Causal relationships must therefore remain versionable.

VCCM Boundary with VCIM

VCIM determines:

whether Value Change occurred.

VCCM determines:

what causal basis is legitimately linked to that change.

Therefore:

Change Identification $\neq$ Causation.

VCCM Boundary with VCTM

VCTM determines:

the type and direction of Value Change.

VCCM determines:

the causal relationship associated with the change.

Therefore:

Change Type $\neq$ Cause.

VCCM Boundary with VCMM

VCMM determines:

how much value changed.

VCCM determines:

what causal basis may explain or contribute to that change.

Therefore:

Magnitude $\neq$ Causation.

VCCM Boundary with VCRM

VCCM establishes the governed causation link.

VCRM preserves the Value Change and its governed characteristics as a lifecycle record.

Therefore:

Causation Determination $\neq$ Value Change Record.

VCCM Boundary with External Domains

VCCM does not replace:

scientific causality methodologies; legal causation tests; insurance coverage determinations;

accounting treatment; tax determinations; regulatory findings; engineering root-cause analysis; medical causation; forensic investigation; financial valuation methodologies.

External systems may establish the causal facts.

VFM® governs:

how those causal facts relate to the governed Value Change.

Minimum Implementation Framework

1. Value Change Input

Receive the governed Value Change from VCIM, VCTM, and VCMM.

2. Candidate Causal Factor Identification

Identify events, conditions, activities, mechanisms, decisions, interactions, or other factors potentially related to the change.

3. Causal Basis Identification

Identify the evidence, methodology, external determination, or mechanism supporting the proposed link.

4. Temporal Relationship Assessment

Determine whether the relevant temporal sequence is consistent with the proposed causal relationship.

5. Mechanism Assessment

Determine whether a plausible or established mechanism connects the factor to Value Change.

6. Alternative Cause Assessment

Identify materially plausible competing or complementary explanations.

7. Multi-Cause Assessment

Determine whether several factors contributed to the Value Change.

8. Contribution Assessment

Where supportable, determine the magnitude or role of individual causal contributions.

9. Causation Link Determination

Assign the appropriate governed causation-link state and strength.

10. Causation Link Record

Document the causal basis, uncertainty, limitations, alternative explanations, contribution, and determination.

Governance Outputs

VCCM may produce:

Value Change Causal Factor Record Value Change Causal Basis Record Value Change Causal Chain Value Change Causal Evidence Record Value Change Alternative Cause Record Value Change Confounding Factor Record Value Change Multi-Cause Map Value Change Causal Contribution Record Value Change Causation Link State Value Change Causation Strength Assessment Value Change Causation Uncertainty Record Value Change Causation Link Record

These outputs provide the governed input for:

VCRM — Value Change Record Module.

Use Case 1 — Banking: Understanding Why

Collateral Value Deteriorated

A bank detects a material reduction in the governed value of an industrial borrower and its production assets.

VCS has already established:

Production Capacity: –18%

Operating Efficiency: –14%

Governed Economic Value: materially reduced.

Several events occurred during the same period:

energy prices increased;

a major customer contract was lost;

production equipment experienced repeated downtime;

interest expense increased.

VCCM prevents the bank from simply declaring:

“the asset lost value because the market deteriorated.”

Instead, it structures the causal relationship:

Customer Loss → reduced production demand → reduced capacity utilization → negative Value Change.

Equipment Downtime → reduced operational output → increased unit cost → negative Value Change.

Energy Price Increase → increased operating cost

→ negative Value Change.

Financing Cost Increase → reduced economic return → negative Value Change.

The bank can then distinguish between:

temporary external pressure,
structural operational deterioration,
and
potentially reversible Value Change.

This materially improves:

credit monitoring, collateral assessment, restructuring decisions, covenant review, impairment analysis, and recovery planning.

Use Case 2 — Space Mission: Identifying the

Source of Mission-Value Degradation

A spacecraft mission experiences a material reduction in expected operational value.

The Value Change has already been identified:

Available Mission Power: –12%

Expected Instrument Operating Time: –17%

Projected Scientific Output: materially reduced.

Several potential causal factors exist:

solar-array degradation;

unexpected thermal conditions;

higher-than-planned subsystem consumption;

battery efficiency loss;

changed mission trajectory.

VCCM structures the causal relationships rather than collapsing the entire Value Change into one explanation.

The organization may establish:

Solar-Array Degradation → lower available generation → strongly supported negative contribution.

Thermal Conditions → increased cooling requirement → supported negative contribution.

Battery Efficiency Loss → reduced usable stored energy → possible additional contribution.

The resulting causation map allows mission planners to distinguish between:

what value has been lost,

why it has been lost,

and

which causal mechanisms may still be operationally changed.

That distinction can directly support:

mission replanning, resource allocation, subsystem prioritization, scientific scheduling, mitigation, and recovery of remaining mission value.

Enterprise Application

VCCM allows organizations to move beyond:

“value changed”

toward:

“this governed Value Change is linked to these identified causal factors, through these mechanisms, with this evidential strength, these contributions, these alternatives, and these uncertainties.”

This capability is directly applicable to:

banking; insurance; investment; industrial production; infrastructure; AI systems; computational systems; energy; blockchain; digital assets; supply chains; organizational capability; intellectual property; mission systems; public assets; technology portfolios.

It is especially valuable where organizations must distinguish between:

symptom and cause,

correlation and causation,

temporary and structural Value Change,

and

single-cause and multi-causal Value Change.

Foundational Principles

Value Change does not establish its own cause.

Temporal sequence is not causation.

Correlation is not causation.

Attribution is not automatically causation.

Magnitude of change is not causal evidence.

Causal-link strength is not causal contribution.

Value Change may have multiple simultaneous causes.

Positive and negative causal effects may coexist.

Alternative explanations must remain visible where materially plausible.

Causal uncertainty must not be converted into artificial certainty.

External causal determinations may be consumed without VFM® replacing the methodologies that produced them.

VFM® governs the relationship between causal basis and Value Change; it does not claim universal ownership of causality itself.

Canonical Closing Statement

Value Change Causation Link Module establishes the causal relationship layer of the Value Change Standard by governing how identified and characterized Value Change is connected to events, conditions, activities, mechanisms, decisions, interactions, transformations, and other materially relevant causal factors. Through governance of candidate causes, causal bases, temporal relationships, mechanisms, direct and indirect links, causal chains, contributing causes, multi-causal structures, interacting and offsetting effects, alternative explanations, confounding factors, causal

evidence, contribution, uncertainty, external determinations, causation-link states, and reassessment, VCCM enables VFM® to preserve why value changed without confusing temporal sequence, correlation, attribution, or assumption with established causation. By separating Value Change identification, type, magnitude, causation, and recording into distinct governance functions, VCCM creates a controlled causal bridge between observed Value Change and its governed lifecycle record. The purpose is not to make VFM® a universal theory of causality, but to ensure that whenever a cause is associated with Value Change, the basis, strength, limits, alternatives, and uncertainty of that relationship remain explicit.

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Governed Links

→ [Value Change Standard \(VCS\)](#)

→ [VFM® Architecture Map](#)

→ [Previous: Value Change Magnitude Module \(VCMM\)](#)

→ [Next: Value Change Record Module \(VCRM\)](#)

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